SR-71 / 600 Procurement List

For the V1.0-P0 small demonstrator. P06 uses the self-built Pico 2 + MPU6050 controller; IM600, IM900, IWT603 and HobbyEagle are not purchased. Accept the power and control sample units first, then close up the airframe. Current budget figures are not vendor quotes.

A. Retained locked-in parts

ID	Qty to buy	Item / search term	Acceptance criteria on arrival	Max onboard mass	Budget cap
P01	3 pcs	Happymodel EX1103 KV11000	2S; 1.5 mm shaft; approx. 13.5 x 15.5 mm; max 4.0 g each; one unit as spare	8.0 g (2 installed)	CNY 330
P02	3 sets	2023R/2023 2-inch CW/CCW prop pairs, 1.5 mm bore	Each set contains CW + CCW; confirm a press fit that does not slip before use; never let glue reach the bearing	<=2.0 g	CNY 90
P03	2 packs	2S 550mAh LiPo, XT30	Max 32 g each; max 60 x 30 x 16 mm; same batch; measure voltage sag; only one pack installed at a time	32 g	CNY 180
P04	1 unit + 2 cells	RadioMaster Pocket ELRS 2.4GHz + 2 x 18650 cells	Confirm the ELRS version; transmitter packaging usually excludes cells; use qualified same-batch 18650 cells	Ground equipment	CNY 650
P05	1 pc	RadioMaster RP1 V2 2.4GHz ELRS	5V, CRSF, max 2.2 g with antenna; verify the UFL antenna is intact	2.2 g	CNY 150
P07	3 pcs	EMAX ES9051 micro digital servos	4.0-5.5 V; max 4.3 g each; approx. 19.8 x 8.9 x 23.1 mm; torque >= 0.8 kg·cm at 4.8 V	8.6 g (2 installed)	CNY 150
P08	1 pc	5V 3A standalone BEC	2S-capable input; max 5 g; must drive the control board and both servos without rebooting; output <= 5.25V	5.0 g	CNY 80

P01/P02 must be power-tested as one matched set; install two P07 units and keep the third as a spare. P08 is raised to 3A; retain an existing 2A BEC only if it holds voltage without reboot while simultaneously driving both servos and the control board.

B. P06 self-built flight-control replacement parts

ID	Qty to buy	Item / search term	Acceptance criteria on arrival	Max onboard mass	Budget cap
P06A	1 board	Raspberry Pi Pico 2, no wireless, no pin headers	Genuine or fully pin-compatible; 51 x 21 mm; UART, I2C and PWM available; board mass <= 4 g	4.0 g	CNY 50
P06B	2 boards	MPU6050 six-axis module with INT pin	Install 1, keep 1 for cross-check/spare; 400kHz I2C; SDA/SCL idle-high level <= 3.3V; max 3 g each	3.0 g (1 installed)	CNY 40
P06C	2 chips	SN74AHCT125N, DIP-14	Must be AHCT125, not HC125; 4 channels; install 1, spare 1	Counted in the 3 g total below	CNY 20
P06D	1 set	Low-ESR capacitors, X7R capacitors and resistors	470uF/10V low-ESR x2; 10uF/25V X7R x5; 0.1uF/100V ceramic x10; 10k ohm x10; 4.7k ohm x10	Counted in the 3 g total below	CNY 30
P06E	2 boards + 1 set	Lightweight perfboard, pin headers/plugs, 26-30AWG signal wire	One board for the aircraft, one for practice/spare; no loose Dupont jumpers in the build; P06A-E target <= 10 g	<= 3.0 g (C-E total)	CNY 30

The MPU6050 supplies raw six-axis data; the Pico handles timestamping, calibration, attitude fusion and PID. The onboard target for P06A-E is <= 10 g; if exceeded, pause close-out and recompute the center of gravity and thrust-to-weight ratio.

C. Purchase after interface confirmation

ID	Qty to buy	Item / spec	Purchase condition	Budget cap
C01	2 pcs	Brushless ESC: 2S, conventional 50Hz PWM, >= 12A continuous each, <= 6 g, <= 25 x 12 x 5 mm	Seller must confirm in writing that conventional receiver PWM works; do not buy DShot-only ESCs; BEC-less preferred	CNY 180
C02	1 set	UHU POR-type foam-safe glue + 30-minute slow-set epoxy	Run corrosion/peel test coupons on same-batch foam first; epoxy only at local spots such as carbon rods and firewalls	CNY 100
C03	1 set	Motor mounting screws/washers	After motors arrive, measure hole spacing, thread type and safe length; screws must not touch the windings	CNY 40
C04	1 set	Phase wires, silicone wire, heat-shrink, XT30, signal wire	Select for the twin 18.4A peak sensitivity; extend phase wires, keep battery DC leads short	CNY 100
C05	As needed	5V to 3.3V unidirectional level shifter	Buy only if the measured RP1 TX high level is > 3.6V; the part must handle the CRSF serial baud rate	CNY 30

Both ESCs use conventional PWM input. Only 3.3V pull-ups are permitted on MPU6050 SDA/SCL; measure the RP1 TX high level before wiring it into the Pico. If an ESC carries a BEC, isolate its red wire; the control bus is powered only by the standalone BEC.

D. Airframe and structural materials

ID	Qty	Material / size	Use and requirements	Budget
M01	2 sheets	3 mm MPP/Depron-type paperless light foam, each sheet >= 650 x 450 mm	Wing/chines, central fuselage, elevons; weigh a 100 x 100 mm sample first, target density about 35-45 kg/m³	CNY 100
M02	1 sheet	2 mm same-type paperless light foam, >= 650 x 400 mm	Roll skins for both nacelles, intake spike webs and finish parts; do not buy 10 mm XPS and split it yourself	CNY 50
M03	1 sheet	1 mm aircraft plywood, >= 300 x 300 mm	Battery tray, bulkheads, nacelle formers, firewall lamination	CNY 100
M04	1 tube	Carbon tube OD4/ID2, >= 500 mm	Cut the 380 mm transverse main spar; must be straight and crack-free	CNY 50
M05	2 rods	OD2 solid carbon rod, >= 300 mm each	Cut a 215 mm longeron from each; a solid rod has no inner diameter	CNY 40
M06	approx. 50 g	PETG 1.75mm	8 printed parts; finished total mass <= 4 g; print one side as a sample first	CNY 60
M07	1 set	0.8 mm steel wire, micro control horns, tape hinges	Left/right pushrods, control horns, elevon hinges	CNY 40
M08	1 set	Hook-and-loop straps, anti-slip pad, belly wear tape, removable steel plates	Dual battery retention; steel plates for CG adjustment, budget at most 16 g	CNY 80

E. Tools and ground equipment

Tool	Minimum capability	Notes
0.01 g scale	Range >= 300 g	Weigh parts and the complete aircraft; a kitchen scale is unsuitable for 4 g parts
Power meter / ammeter	2S, >= 30 A	Record voltage, current and voltage sag
Thrust stand	Range >= 500 gf, with prop guard	Can be homemade, but the motor and scale must be rigidly fixed
Multimeter	DC voltage, continuity	Verify the 5V rail and polarity
Balance charger	2S capable, balance/storage modes	Do not use simple USB charge leads without balance management
Square / calipers / angle gauge	0.1 mm class calipers	Verify hole spacing, axes, 15° fin cant and 3° initial elevon reflex
Low-wattage iron / hot air or lighter	Suitable for fine-wire soldering	Provide strain relief at joints; keep heat away from foam when shrinking
Adjustable current-limited PSU	0-12 V, >= 3 A	Current-limit the control board at first power-up to avoid cascading damage from wiring errors
USB logic analyzer	>= 8 channels, >= 24 MHz	Check I2C, CRSF serial, PWM and control loop timing
Goggles and prop guard	Covers test personnel	Nobody stands in the prop disc plane during powered bench tests

F. Not purchased for now

[ ] 2.5-inch propellers: they change current draw and prop-disc clearance; re-match only if the 2-inch setup measures insufficient.

[ ] Parachute, release servo and shroud lines: only a space reservation exists so far, with no deployment-load or altitude validation.

[ ] Landing gear, video link, autopilot equipment and decorative lights: they consume mass and CG margin.

[ ] Whole-airframe 3D-print material: this design is foam-based; a fully printed shell would break the mass budget.

[ ] Final motor hole template: transfer it only after the actual motors arrive; the current W03 is a blank.

G. Incoming acceptance checklist

[ ] Weigh every item and enter measured values into data/bom.csv; never substitute vendor nominal masses.

[ ] Measure motor diameter, length, shaft diameter, hole spacing, thread and usable screw length with calipers.

[ ] Confirm CW/CCW prop bore, thrust direction and press-fit security; check rotation direction powered, without props installed.

[ ] Measure each battery pack for dimensions, mass, rest voltage and internal resistance/voltage sag; retire it immediately on swelling, damaged wrapper or an abnormal cell.

[ ] With the BEC feeding the control board and both servos, run simultaneous rapid-motion/light-load tests; the 5V bus must not sag or reboot.

[ ] Bench-verify each ESC individually for PWM, throttle calibration, temperature rise and protection before pairing them into the twin installation.

[ ] Repeatedly flash and power-cycle the Pico over USB; connect the RP1 to UART0 and the MPU6050 to I2C0 and INT, and log continuously for 10 minutes without dropped samples.

[ ] Measure the SDA/SCL idle-high levels and the RP1 TX high level; Pico GPIO must never directly withstand more than 3.6V.

H. Compatibility review against original equipment

Original equipment / interface	New-plan correspondence	Check conclusion
P04 Pocket ELRS -> P05 RP1	ELRS radio link unchanged	Match
P05 RP1, 5V/CRSF	RP1 TX -> Pico UART0 RX; common ground	Protocol match; direct wiring only after the TX high level measures <= 3.6V
P06B MPU6050	SDA->GP4, SCL->GP5, INT->GP3; I2C 400kHz	Function match; Pico timestamps on DATA_RDY and runs its own attitude fusion
P06A Pico 2	UART0 to RP1; I2C0 to MPU6050; 4 PWM channels to AHCT125	Sufficient resources
P07 ES9051, two units	AHCT125 5V PWM -> left/right servos	Voltage match; start at 50Hz; loaded test mandatory
C01 two conventional PWM ESCs	AHCT125 5V PWM -> left/right ESCs	Match; DShot-only ESCs do not match
P08 BEC	5V/3A -> RP1, Pico VSYS, AHCT125 and servos	Match; an existing 2A version must pass the simultaneous load test
Front-bay installation space	Pico 51 x 21 mm, MPU6050 module per actual size, stacked installation X approx. 295-351	Original A3 envelope void; dry-fit assembly before cutting the tray
Mass / CG	Planned mass 197.9 g, CG X=296.4 mm; CG stays inside the window with the battery at X=120	10 g control-stack target is feasible; recompute after weighing
Thrust margin	Manufacturer peak static thrust-to-weight ratio about 1.23; about 0.99 assuming 20% installation loss	Margin is thin; no powered flight if the measured installed static thrust-to-weight ratio is < 1.2

Budget: the current project table totals about CNY 3230, which is CNY 0 above the 3500 first-round target; still below the 5000 overall cap, with about CNY 1770 of headroom. Prices vary by channel; passing the specification and mass checks takes priority.

Specification references

Happymodel EX1103 KV11000: https://www.happymodel.cn/index.php/2022/09/05/bassline-spare-part-ex1103-kv11000-brushless-motor/
RadioMaster RP1 V2: https://www.radiomasterrc.com/products/rp1-expresslrs-2-4ghz-nano-receiver
Raspberry Pi Pico 2: https://www.raspberrypi.com/products/raspberry-pi-pico-2/
MPU6050 official datasheet: https://invensense.tdk.com/wp-content/uploads/2015/02/MPU-6000-Datasheet.pdf
TI SN74AHCT125: https://www.ti.com/product/SN74AHCT125
EMAX ES9051: https://shop.emax-usa.com/products/emx-sv-0272-es9051